Hierarchical Attention-based Reconciliation of Multimodal Ontologies for Image Creation

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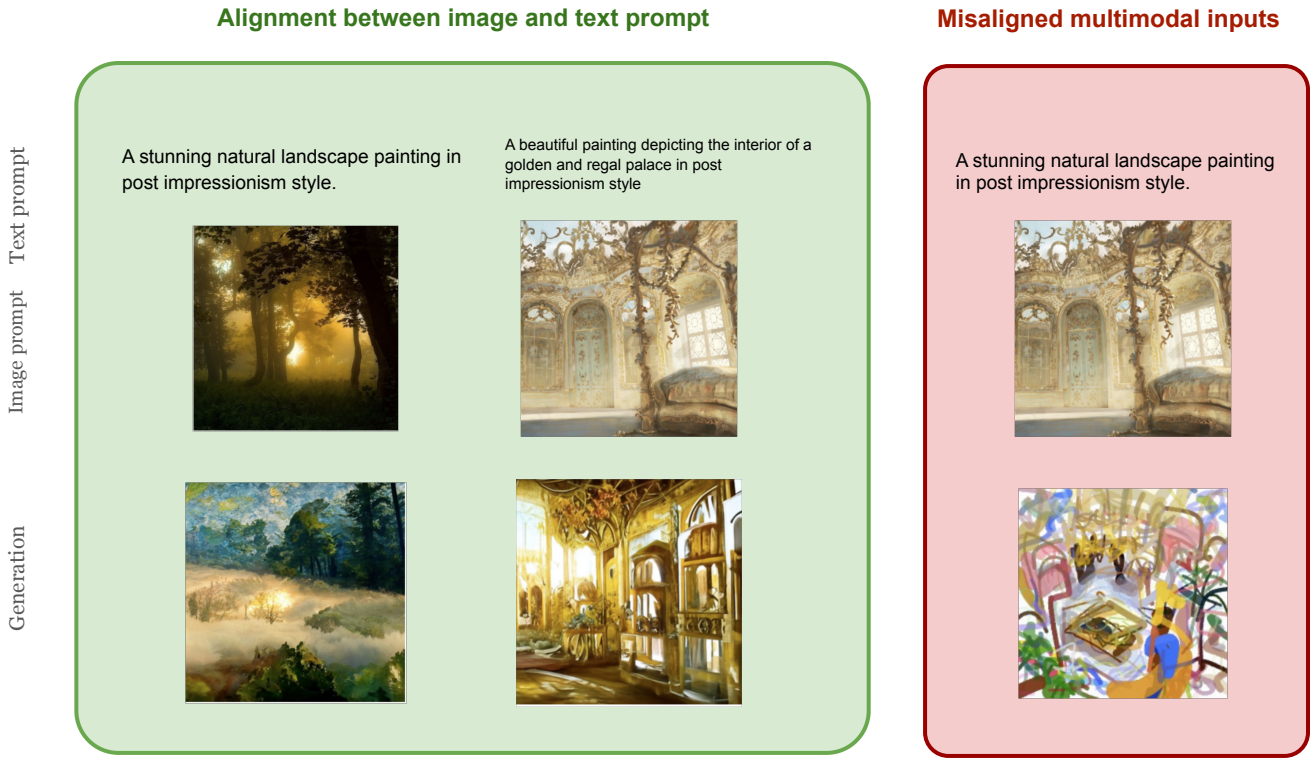
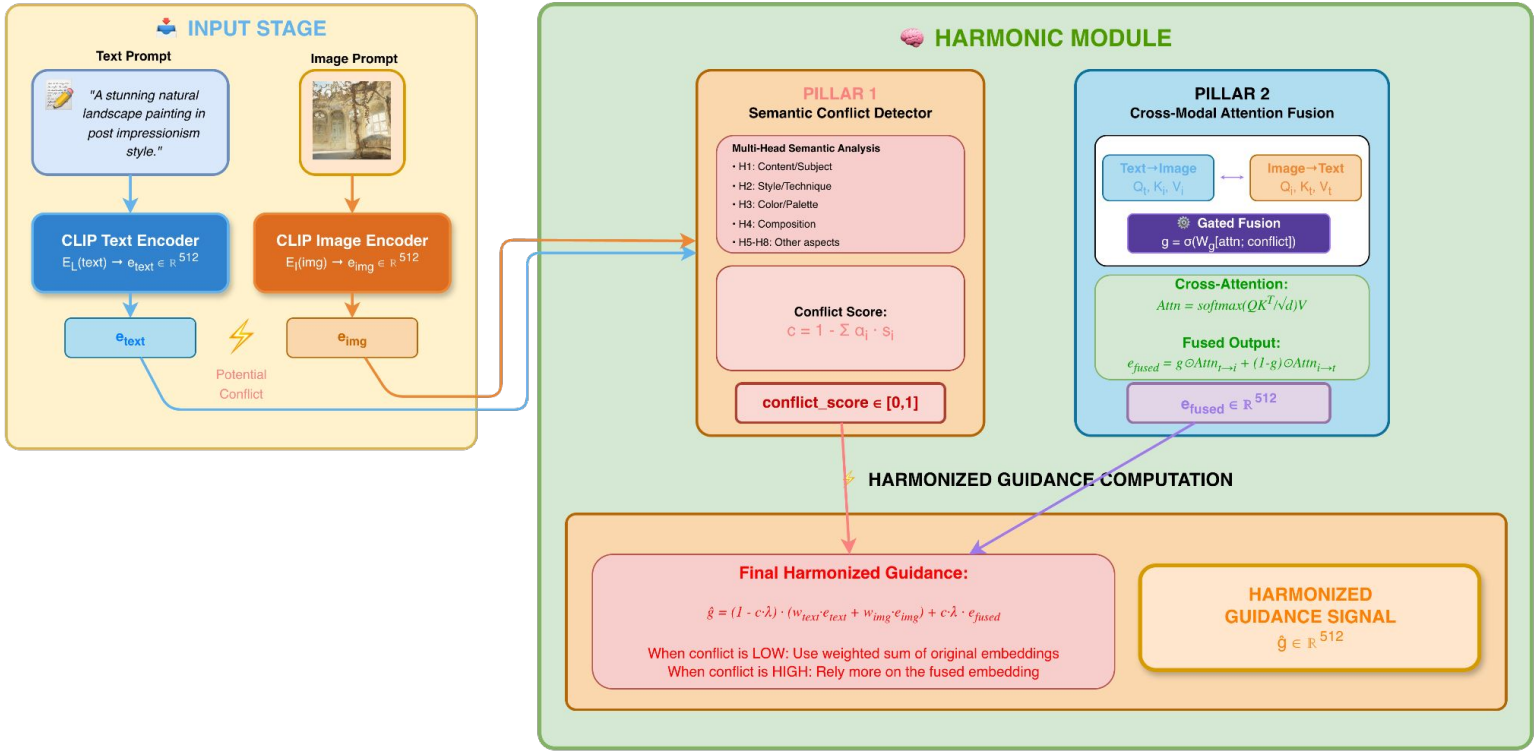
Motivation



Low semantic similarity leads to poor generation results

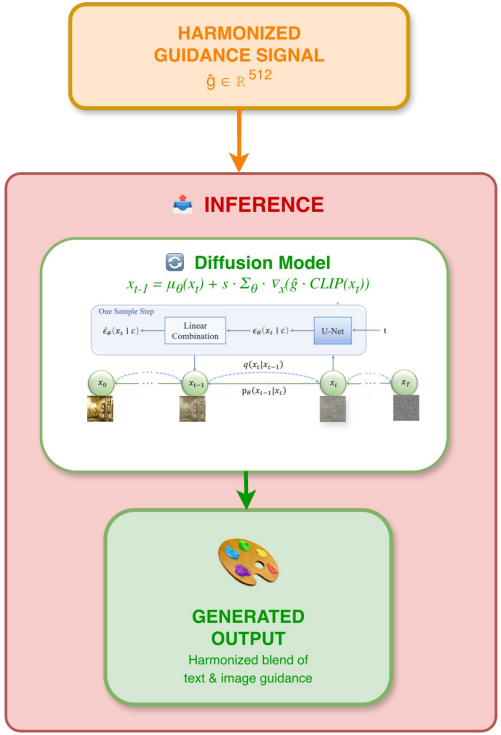
If the difference between image prompt and text prompt in semantic content is too large, then it will lead to the generation of unsatisfactory results. We propose the adaptive prompt discarding module, which calculates the similarity between image prompt and text prompt and then compares the weights of the two to decide whether one of the modalities should be discarded to obtain a better generation result

Architecture



Instead of discarding modalities, HARMONIC extracts the INTERSECTION of compatible semantic concepts from BOTH modalities, enabling creative blending even with conflicting prompts.

Architecture



Harmonized signals are used at inference

Pillar I - Semantic conflict detection - CLIP based similarity and conflict scoring

Pillar II - Cross-modal attention fusion - Soft attention over features. The Cross-Modal Attention Fusion module identifies and extracts complementary information from both modalities.